

Executive summary: mortgage PD model validation

Decision: approved with conditions, for rank-ordering in benign conditions only. Not approved for absolute PD, provisioning or capital use.

Both models rank borrowers well (hold-out AUC 0.84 scorecard, 0.85 LightGBM). Out of time they keep ranking reasonably (AUC about 0.78), but their predicted default rates are far too low.

Top three findings: 1. **Severe under-prediction in the crisis (F1, High)**. For 2006-08 originations the models predict 1.1-1.1% against 4.7% observed (4.4x scorecard, 4.1x LightGBM). All 10 of 10 score bands are under-predicted. In 2005 the gap is already 2.1x and 1.9x. 2. **The error is concentrated (F2, High)**. California loans are under-predicted 14x, Nevada 11x, Arizona 9x and Florida 8x, and broker-channel loans 10x. One margin of conservatism would be too much for some segments and too little for others. 3. **Input drift (F3, High)**. Interest rate PSI is 1.4 in the crisis and 6.6 in 2016-19, and the scorecard's own score distribution shifts sharply in 2016-19 (PSI 0.52). In that period the scorecard under-predicts 1.8x and LightGBM 1.26x.

Conditions: (a) use only to rank or tier, not to estimate PD; (b) apply a conservatism margin or recalibrate on stressed data before any PD use; (c) monitor PSI and calibration quarterly (Section 10); (d) redevelop if the triggers in Section 10 fire.